SMU SINGAPORE MANAGEMENT UNIVERSITY MASTERS

Global Summer Programme
ACCT682
Programming for Business Analytics
Dr Wang Jiwei

TEAM PROJECT INSTRUCTIONS

Aim

The group project aims to provide students with an understanding of programming which helps to draw insights from business data in forecasting and forensics. Students should be able to use statistical and machine learning algorithms and other analytics tools to critically analyze and evaluate various aspects of a given topic and dataset. The research topics are presented in the Appendix.

Composition of teams

The number of team members depends on the size of your enrolled class section. All group members should be registered for the same class section. I reserve the right in my ultimate discretion to allocate group members.

Deadline

Deadline for ***softcopy*** submission of the project report with code in Jupyter Notebook (.ipynb) format, data, and presentation slides is ***1159pm, Wednesday, 05 Aug 2026***. No extension of the deadline date will be entertained. Early submission is welcomed, although no bonus marks will be awarded.

Online Submission Instruction

You are required to submit all your files, including but not limited to *project report with code*, *data*, and *presentation slides*, through SMU official OneDrive. Please do not zip files or folders and you should share the folder and files with me directly. You are advised to perform the following steps to share files through OneDrive:

1. Download the OneDrive app to your local computer.
2. Copy the folder with all submission files to the OneDrive folder on your local computer. It will automatically synchronize to the cloud drive. You may want to run all the code again to make sure all are ready.
3. Right click on the OneDrive folder on your local computer and then "Share" with my email address jwwang@smu.edu.sg. Make sure to change the share setting to "People you specify can edit."
4. Click "Send" to share the folder with me.

You are required to use the Jupyter Notebook file format (.ipynb) which contains both your report and code. The cover page of the report should clearly state the title of the research report, the names (**as your matriculation names and in alphabetical**

order) and Student ID numbers of all the team members, class section, the date of submission (month & year will suffice), and the instructor's name. The file name should follow the prescribed format: coursecode_groupnumber_topic (e.g., ACCT682_G1_FraudDetection.ipynb).

You are advised to use relative file path wherever necessary; thus I can reproduce your code without changing any code. I will not grade the project if I cannot reproduce your code.

If any part of your code runs more than 3 minutes on your computer, please clearly state it at the beginning of your Jupyter Notebook file for me to take note of.

Project outline

Your written report should include, but not be restricted to, the following:

1. An introduction section to present the business problem and a preview of the main points of the in-depth report. The introduction should contain enough information for a reader to get familiarized with what is discussed in the full report without having to read it.
2. Exploratory data analysis including but not limited to an in-depth look at the main data features (dependent variable and some main independent variables), missing values, outliers, categorical variables, distributions, and correlations of the dataset. If you are using external data, clearly state the source of the data and how you obtain the data.
3. Model selection, implementation, evaluation, and refinement: explain your reasoning for model selection. Train your models using the training dataset and test your models using the testing dataset. Evaluate your models using the metrics provided for the topic. You may use any statistical algorithms introduced in this course and other courses. Explain the process your group took for refining/iterating on your model.
4. Conclusion: how well did your model do? What lessons did you learn from this project? Make a conclusion to close your project report. It is recommended to end the report with an appropriate, meaningful final sentence that ties the whole point of the report together.

Presentation

Each group is required to give a *30-minute presentation* (including Q&A and class discussion) in class. All group members are required to take part in the presentation. Presentation order will be randomly scheduled in the last session.

Assessment & Evaluation

The team project report carries 50% of the total marks for the course. Credit will be given for good analysis, quality content and rationale, prediction accuracy, creativity, good writing and programming style, insightful report, and an interesting oral presentation.

Academic Integrity

All work is to be performed exclusively by the members of the team and all team members must contribute their fair share to the project. Teams are not to consult with other teams, nor to obtain help from any other persons within or outside of SMU. If outside research is used, sources are to be cited and information discovered via outside research is to be clearly labeled as such (in appropriate footnotes or in a bibliography appendix). If outside research is performed, the product of your research is not to be shared with any student who is not a member of the team.

Unless I hear from you otherwise, I will assume that all members of your team are contributing their fair share to the project and I will award the same marks to all members of your team. If you have strong grounds to contend that a particular member of your team should not receive the same marks as the other member(s) including yourself, you should send an email to me independently under confidential cover explaining clearly the reasons for your allegation not later than the deadline date for the submission of your team report. I will evaluate each allegation on its own merit, but my ultimate decision with respect to the award of marks to each team member shall be final.

I view very seriously any plagiarism of previous terms' project reports done by other students. The penalty for such plagiarism is an "F" grade for the whole course.

AI applications such as ChatGPT and Claude are allowed to assist students on coding but students are prohibited from copying codes from AI applications or using the applications to write report.

Appendix A Project Topics

A selected project topic from the following will be pre-assigned to your group on eLearn. Check eLearn to confirm your project topic.

Topic 1: Bank Churn
This is a past Kaggle competition. Details are available at:
https://www.kaggle.com/competitions/playground-series-s4e1/

Topic 2: Loan Approval
This is an ongoing Kaggle competition. Details are available at:
https://www.kaggle.com/competitions/playground-series-s4e10/

Topic 3: House Prices
This is a past Kaggle competition. Details are available at:
https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/

Topic 4: Store Sales
This is a past Kaggle competition. Details are available at:
https://www.kaggle.com/competitions/store-sales-time-series-forecasting/

--END--